

GIT Server Setup

Step-1: Install git Package

```
# sudo apt install git -y
```

```
siddharrth@master:~$ sudo apt install git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme at-spi2-common at-spi2-core dconf-gsettings-backend dconf-service fontconfig
  gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme javascript-common
  libatk-bridge2.0-0t64 libatk1.0-0t64 libatspi2.0-0t64 libcairo-gobject2 libcairo2 libcolord2 libdatriel libdconf1
  libepoxy0 libexpat1-dev libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgraphite2-3
  libgtk-3-0t64 libgtk-3-bin libgtk-3-common libharfbuzz0b libjs-jquery libjs-sphinxdoc libjs-underscore liblcms2-2
  libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpixmap-1-0 libpython3-dev libpython3.12-dev librsvg2-2
  librsvg2-common libthai-data libthai0 libwayland-client0 libwayland-cursor0 libwayland-egl1 libxcb-render0
  libxcb-shm0 libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxi6 libxinerama1 libxrandr2 libxrender1 libxtst6
  python3-dev python3-wheel python3.12-dev session-migration ubuntu-mono x11-common zlib1g-dev
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

```
# git --version
```

```
siddharrth@master:~$ git --version
git version 2.43.0
```

Step-2: Create a Git User

```
# sudo useradd git
```

```
# sudo passwd git
```

```
siddharrth@master:~$ sudo useradd git
siddharrth@master:~$ cat /etc/passwd | tail
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uuidd:x:104:105::/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
siddharrth:x:1000:1000:siddharrth:/home/siddharrth:/bin/bash
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
git:x:1001:1001::/home/git:/bin/sh
siddharrth@master:~$ su - git
Password:
su: Authentication failure
siddharrth@master:~$ sudo passwd git
New password:
Retype new password:
passwd: password updated successfully
```

Step-3: Create a Bare Repository

```
# sudo su - git
# mkdir project.git
# cd project.git
# git init --bare
```

```
$ mkdir project.git
$ cd project.git
$ pwd
/home/git/project.git
$ git init --bare
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/git/project.git/
```

Step-4: Setup SSH Access

```
# ssh-keygen
```

```
Initialized empty Git repository in /home/git/project.git/
$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/git/.ssh/id_ed25519):
Created directory '/home/git/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/git/.ssh/id_ed25519
Your public key has been saved in /home/git/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:ylT7yJ54bPceR434sNNra0hLIZK+4Mm9tNkcuIMw0Y8 git@master
The key's randomart image is:
+--[ED25519 256]--+
|
| . .
| . . o . o
| ..oS . + o .
| ooE+. = . B
| o+o* o.*.+
| o.B=Oo+o+..
| =oX=++o..
+-----[SHA256]-----+
```

Step-: Clone Repository from Client

```
siddharrth@master:~$ git clone git@10.10.40.88:/home/git/project.git
Cloning into 'project'...
The authenticity of host '10.10.40.88 (10.10.40.88)' can't be established.
ED25519 key fingerprint is SHA256:qIAY3K9tMCLZxyD0z3i4MdEGmby+Pqroxpo3i/pYy1g.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.40.88' (ED25519) to the list of known hosts.
git@10.10.40.88's password:
warning: You appear to have cloned an empty repository.
siddharrth@master:~$ ls
project  svr  wget-log
```

```
siddharrth@master:~$ cd project/
siddharrth@master:~/project$ echo "hello git" > index.file
siddharrth@master:~/project$ cd
siddharrth@master:~$ cd project/
siddharrth@master:~/project$ git add .
siddharrth@master:~/project$ git commit -m "first commit"
[master (root-commit) 02b8777] first commit
Committer: siddharrth <siddharrth@master.smb.com>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:
```

```
git config --global --edit
```

After doing this, you may fix the identity used for this commit with:

```
git commit --amend --reset-author
```

```
1 file changed, 1 insertion(+)
create mode 100644 index.file
```

```
siddharrth@master:~/project$ git push origin master
git@10.10.40.88's password:
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 218 bytes | 218.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To 10.10.40.88:/home/git/project.git
* [new branch]      master -> master
```

```
sri@master:~$ git clone git@10.10.40.88:/home/git/project.git
Cloning into 'project'...
The authenticity of host '10.10.40.88 (10.10.40.88)' can't be established.
ED25519 key fingerprint is SHA256:qIAY3K9tMCLZXyD0z3i4MdEGmby+Pqroxpo3i/pYy1g.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.40.88' (ED25519) to the list of known hosts.
git@10.10.40.88's password:
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
Receiving objects: 100% (3/3), done.
sri@master:~$ ls
project
sri@master:~$ cd project/
sri@master:~/project$ ls
index.file
sri@master:~/project$ cat index.file
hello git
```

Reference:

<https://git-scm.com/book/en/v2/Git-on-the-Server-Setting-Up-the-Server>